

Advanced Regression Model Comparison Report

Project : House Price Prediction
Dataset : House Prices
Target : SalePrice

Executive Summary

Seven regression algorithms were trained and evaluated using R² Score, MAE, MSE, RMSE and Cross Validation. The best performing model is highlighted below.

Model Performance

Model	R ² Score	MAE	MSE	RMSE	CV Score
CatBoost	0.9280	14,311.23	397,771,067.49	19,944.20	0.9098
XGBoost	0.9204	14,809.86	439,669,632.00	20,968.30	0.9057
Gradient Boosting	0.9155	15,051.90	466,691,468.39	21,603.04	0.9023
LightGBM	0.9095	15,441.47	499,742,861.74	22,354.93	0.9071
Random Forest	0.8902	16,336.49	606,723,422.48	24,631.76	0.8786
Linear Regression	0.8428	20,520.77	868,427,623.65	29,469.10	0.9030
Decision Tree	0.8236	23,058.13	974,621,077.63	31,218.92	0.7791

Best Model

Model : CatBoost
R² Score : 0.9280
MAE : 14311.23
RMSE : 19944.20
CV Score : 0.9098

Conclusion

Based on the evaluation metrics, **CatBoost** achieved the best prediction performance and is recommended for deployment.